Lesson Planning

Complete Lesson Plan Book

Class: Class 12

Subject: Physics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Current Electricity: Ohm's Law

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Learning Objectives

• Students will state Ohm's law and write its formula V = IR.

• Students will calculate current, voltage or resistance from given values.

• Students will explain why a filament bulb does not obey Ohm's law perfectly.

Key Concepts

• Ohm's law states that the current through a conductor is directly proportional to the voltage across it, provided temperature stays constant.

• The law is written as V = IR, where V is voltage in volts, I is current in amperes and R is resistance in ohms.

• Resistance measures how much a conductor opposes current; a longer or thinner wire has higher resistance.

• A metal wire at steady temperature obeys Ohm's law, but a filament bulb heats up, so its resistance rises and its graph curves.

Teaching Methodology (Step by Step)

1. Ask what happens to a bulb's brightness when a second cell is added, and link brightness to current.

2. Introduce Ohm's law with a simple circuit diagram and the formula V = IR.

3. Solve three calculations on the board: find I, find V, find R.

4. Compare the straight-line graph of a wire with the curved graph of a bulb.

5. Students solve five Ohm's law problems and check answers in pairs.

Classroom Activities

• Circuit prediction: groups predict ammeter readings before changing voltage.

• Graph reading: students identify ohmic and non-ohmic conductors from graphs.

• Formula triangle: students practise covering V, I or R to recall the formula.

Assessment Questions

1. State Ohm's law in your own words and write its formula.

2. A 12 V battery drives 2 A through a wire. What is the resistance?

3. Why is a filament bulb called a non-ohmic conductor?

Homework

• Solve: V = 6 V, R = 3 ohm. Find I. Then double V and find the new I.

• Explain with a graph why a metal wire obeys Ohm's law but a bulb does not.

Time Allocation

40 minutes